Queen Mary
University of London

EBU5376
Digital Signal Processing

Week 3 – Tutorial 2

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Question 1a

Use the DTFT to derive the frequency response for the system represented by the difference equation:

$$y[n] = 0.5x[n] + 0.5x[n-1]$$

Question 1b

Write an expression for each of the magnitude and phase of the frequency response of the system whose difference equation is:

$$y[n] = 0.5x[n] + 0.5x[n-1]$$

Then calculate the Magnitude and Phase for $\omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ rads

Question 2a

Use z-transforms to derive the frequency response for the system represented by the difference equation:

$$y[n] = x[n] - 0.5y[n-1]$$

Question 2b

Write an expression for each of the magnitude and phase of the frequency response of the system whose difference equation is:

$$y[n] = x[n] - 0.5y[n-1]$$

Then calculate the Magnitude and Phase for $\omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ rads

Question 3

A system's "Characteristic Equation" is the denominator of its transfer function equated to 0.

Determine the stability of the sampled data systems whose characteristic equations are:

a) z^2-0.25=0

b) z^2-z+0.25=0

c) z^2-2z+1=0

d) $z^2 + 1.414$z+1=0

Question 4a

Find the transfer function of the system whose difference equation is:

$$y[n] = \frac{1}{3}\left(y[n-1] - y[n-2] + x[n-1] + x[n-2]\right)$$

Hence determine the system's stability.

Question 4b

Determine the stability of the system whose transfer function is:

$$H(z) = \frac{z^{-1} + z^{-2}}{3 - z^{-1} + z^{-2}}$$

Zero Order Hold

When we introduced sampled systems and referred to digital-to-analog conversion (DAC), we introduced the concept of a Zero-Order Hold (ZOH). This circuit holds the value of a sample till the next sample is taken.

Question 5

Using Laplace Transforms, determine the Transfer Function of a ZOH.

Question 6

The transfer function of a ZOH is G(s) = $\frac{1-e^{-Ts}}{s}$

Determine the magnitude and phase of its frequency response function.

Question 7

Given the pole-zero plot of a system:

- Zeros at $z = 1$ and $z = -1$
- Poles at $z = 0.9e^{-j\frac{\pi}{4}}$ and $z = 0.9e^{j\frac{\pi}{4}}$

Sketch the approximate amplitude response and explain the behavior ~~(e.g., is it a low-pass, high-pass, band-pass, or band-stop filter?).~~

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